MILITARY SPECIFICATION

Cores for Search Coil Assemblies

This amendment forms a part of Military Specification MIL-C-23331(OS) of 11 June 1962 and has been approved by the Naval Ordnance Systems Command, Department of the Navy.

Page 1
By this amendment, Bureau of Naval Weapons is changed to Naval Ordnance Systems Command on page 1 and throughout the specification. On each page change MIL-C-23331(WEP) to MIL-C-23331(OS).

Page 5
Paragraph 4.1.1, second line, change: "...sample of 10 cores..."
To: "...sample of cores..."

Page 6
Paragraph 4.3.1.1 (a) change to read: "4.3.1.1 a. Search Coil Test Solenoid Coil Assembly, Drawing 3212308."
Delete: 4.3.1.1 (b) in its entirety
Add:

b. Recorder, ink writing, strip chart indicating:

(1) Input: D-C voltages (300 v max. upper scale limit)
D-C currents (with external dropping resistor).
(2) Calibrated Accuracy: Span: ±0.25% of span or 1 μv, whichever is greater.
Zero Position: ±(0.25 +.1 X suppression ratio) % of span or 1 microvolt, whichever is greater.
(3) Dead Band: 0.15% of span (0.3% on 0.1 mv span).
(4) Span Step Response Time: Less than 0.5 second. Same for 50 or 60 cycle operation.
(5) Frequency Response: Down less than 1% of full scale for a 5 cps, 10% of full scale peak-to-peak sine wave input.
(6) D-C Voltage Spans: 19 discrete calibrated full scale steps of 0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 50, and 100 millivolts and 0.2, 0.5, 1, 2, 5, 10, 20, 50, and 100 volts.

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(7) Zero Positioning: + 100% to -1000% of span. Rotary switch, 11 fixed zero positions with continuous vernier adjustment to intermediate positions.
(8) Voltage Reference: Double Zener diode bridge.
(9) Input Impedance: Potentiometric on millivolt range - Infinite at balance, minimum off balance impedance of 25,000 ohms (a.c or d.c) per millivolt of span, 1 megohm on 100 mv span. Voltage spans have constant 4 megohms regardless of balance position.
(10) Source Impedance: Maximum source impedance for specified performance - Millivolt spans - 25,000 ohms per millivolt of span; Voltage spans - 10,000 ohms for 0.25% loading error regardless of span.
(11) Damping: Adjustable to prevent overshoot on full scale or partial scale step functions.
(12) Chart and Scale: 6" calibrated width. Evenly graduated in percent from -2 to +102 reading from right to left. Paper continuously ruled with time lines in tenths of an inch; 122 feet per roll. Paper can be continuously rerolled on take-up reel or driven out bottom of chart platen behind lower guide bar. Adjustable sloping chart platen, removeable chart transport for replacing charts.
(13) Chart Speeds: 10 fixed speeds of 10, 5, 2, 1 minutes per inch and 40, 20, 10, 5, 2, 1 seconds per inch plus "Stand By."
Note: Honeywell Model No. 19311-002-000-01, Electronic 193 Lab Recorder, has met the foregoing requirements.

Subparagraph (c), change to read: c. Transformer Mark 17 Mod 0 or 1.
Subparagraph (d), delete in its entirety and reletter remainder of subparagraphs accordingly without parenthesis.

d. Fixed Resistors:
One each (R1) 10 ohms, 3%, 5 watt
(R2) 100 ohms, 3%, 5 watt

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Page 6, paragraph 4.3.1.1, continued

e. Rheostats:
One each (R3) 0-5 ohms, 3 turns, 3 watt
(R4) 0-2000 ohms, 10 turns, 3 watt

f. Milliammeters:
One each (M1) capable of reading 450 ±2 milliampers (ma), calibrated at 450 ma.
One each (M2) having two ranges with the following capabilities:

Low Range - 0-10 ma, accuracy ± 0.05 ma, calibrated at 5.00 ma.
High Range - 10-100 ma, accuracy ± 0.5 ma, calibrated at 30.0 ma.

g. Single Throw Switch
One each (S1) - must carry 5 amperes inductive current at 6 volts; contact resistance not greater than 0.01 ohm.

h. Current Source - one 6-volt storage battery

Page 7

Paragraph 4.4.3.2, after number, add: (M101)

In all subparagraphs, delete parenthesis from subparagraph letters and add a period after the letters.

In subparagraph a., second line, change: 4.3.1.1 (a)
To: 4.3.1.1 a.

In subparagraph b., first line, change: 4.3.1.1 (c)
To: 4.3.1.1 c.

Ninth line, change: 4.3.1.1 (b)
To: 4.3.1.1 b.

Eleventh line, change: 4.3.1.1 (i)
To: 4.3.1.1 h.

Delete: fluxmeter (2 places)
Add: recorder (2 places)

In subparagraph c., seventh line, change: 4.3.1.1 (h)
To: 4.3.1.1 g.

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Page 8, Paragraph 4.4.3.2 continued,
Subparagraph c., tenth line, change: 4.3.1.1 (i)
To: 4.3.1.1 h.
Subparagraph d., change to read:
"The secondaries of the solenoid and the transformer shall be connected in series to the recorder. This connection shall be made in such a way that when the primary circuit is opened, the transformer tends to give a recorder deflection to the left and the solenoid tends to give an opposite recorder deflection to the right. Correct polarity can be established by energizing each primary separately, in turn."

Page 9
Delete Figure 2 and add new Figure 2, Page 5, this amendment.

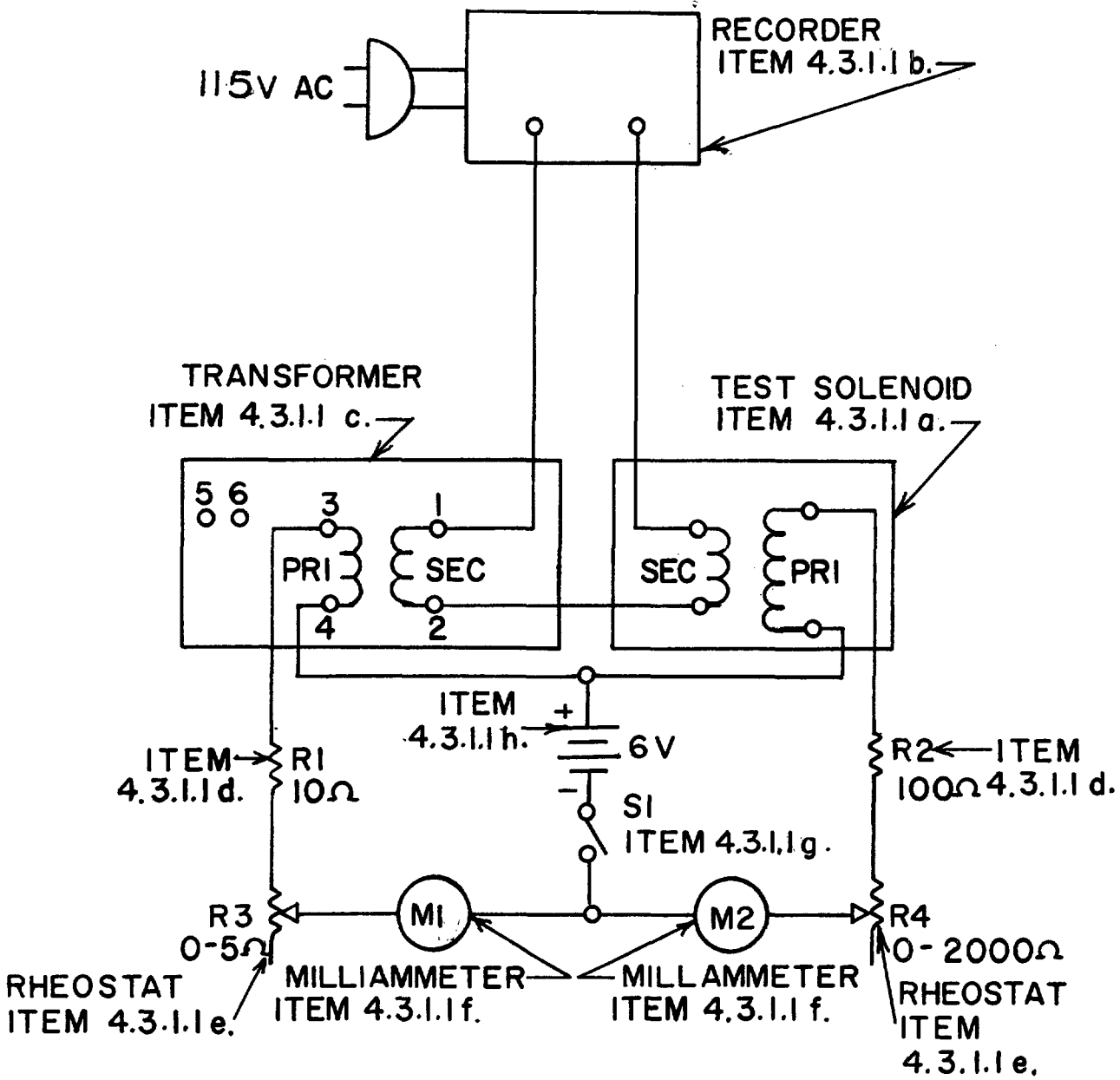
Page 10, Paragraph 4.4.3.2 continued,
Subparagraph g., change to read: "With the recorder off, center the recorder pen."
Subparagraph h., change to read:
"Make the current for a period of a few seconds. Hold sufficiently long for the recorder pen to settle down to the zero center position. Break the current."
Subparagraph i., change to read:
"If the area under the curve, after the switch is opened, is greater to the left of the zero position, the core has failed to meet the requirement of 3.2.2.2. If the area under the curve, after the switch is opened, is greater to the right of zero or the left and right areas are equal, the core acceptable. (The 95 percent factor has already been allowed for in the current values specified in Step e. above.)"
Paragraph 4.4.3.3, after number, add: (M102)

Page 11
Paragraph 4.4.4, after number, add: (M103)

Custodian:
NAVY-OS

Preparing Activity:
NAVY-OS
(Project No. 1350-N028)

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CIRCUIT FOR PERMEABILITY TEST OF CORES

FIGURE 2

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